

Feature Reduction in Time Series CNC Machine Data

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Objective: To reduce high-dimensional CNC (Computer Numerical Control) Machine sensor data to a compact feature set to **retire redundant sensors and cut computational cost.**

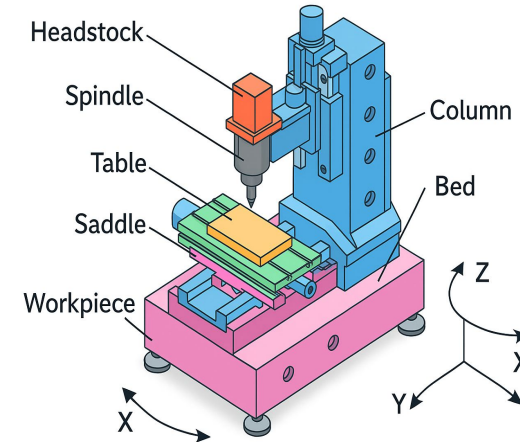
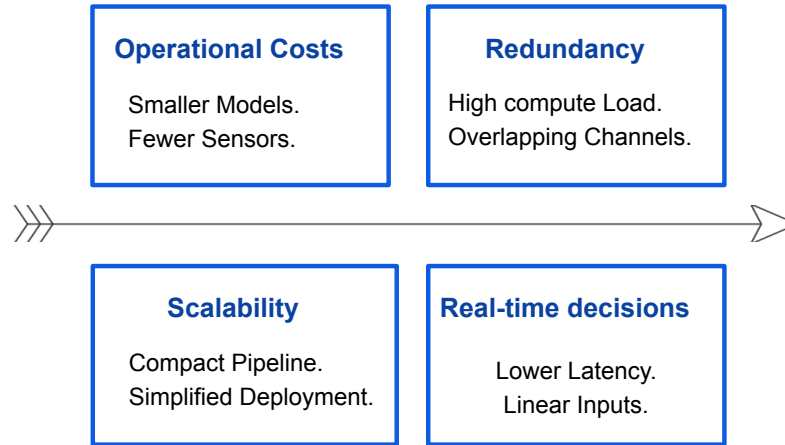


Fig 1: CNC Machine Overview^[7]

Understanding Data

Dataset

Size: (42,016, 52)

Data Type: Numerical, Multivariate time-series data

Components: Steel (S)

Features: LOAD|6, ENC_POS|1, DES_POS|1, TORQUE_FFW|1, CTRL_DIFF|2

Targets: CURRENT|6 (Spindle) , CURRENT|1 (X), CURRENT|2 (Y), CURRENT|3 (Z)

LOAD 1	LOAD 2	LOAD 3	LOAD 6	ENC_POS 1	ENC_POS 2	ENC_POS 6	CTRL_DIFF2 1	DES_POS 1	TORQUE_FFW 1	CURRENT 6	...
0.341797	6.47583	0.421143	0	22.200148	284.270207	181.730994	-0.00001	22.200139	0	-0.273438	...
0.341797	6.481934	0.415039	0	22.200139	284.270197	181.730994	0	22.200139	0	-0.273438	...
0.341797	6.481934	0.415039	0	22.200139	284.270197	181.730994	0	22.200139	0	-0.273438	...
...	...	...	...	...	...	...	...	...	...	...	...

Table 1: DMC_S_CP2 Dataset

Reducing High-Dimensional Data to Compressed Meaningful Features

Feature Reduction Workflow

1. Data Scaling and Splitting

StandardScaler

Training Set = 60%

Validation Set = 20%

Test Set = 20%

2. Data Sequencing

Sliding Windows

Important to learn
temporal features

Depends on the model

3. Hyperparameter Configuration

Latent_Dim = 25 or 14

Window Size = 60 or
120

4. Model Training

Monitor Training Loss

Early Stopping

5. Output

Reduced Features

Identifying the Best Architecture for Feature Selection

Architecture Selection

Principal Component Analysis

Linear
Variance-maximizing
Orthogonal
Components

Autoencoder

Non-Linear Neural
Encoder-Decoder
Network

LSTM+CNN Autoencoder

Hybrid
Temporal-Spatial
Encoder-Decoder
Network

PCA for Feature Reduction

Principal Component Analysis (PCA)

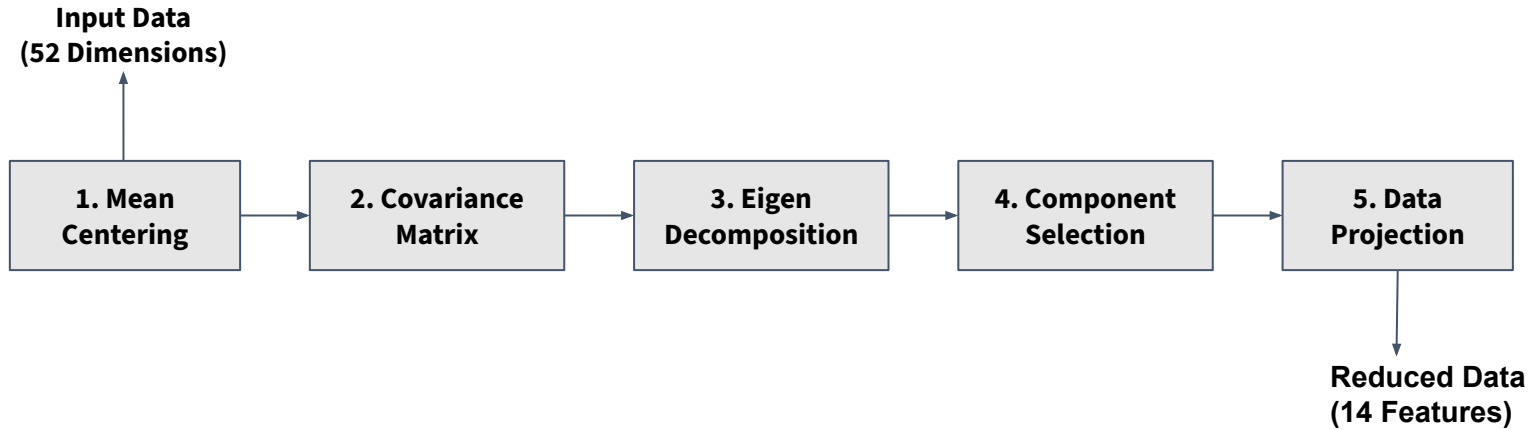
1. Simple and fast

2. Deals with
multicollinearity

3. Strong
Compression

4. Interpretable

5. Reproducible



Hyperparameter	Value
Scaler	StandardScaler
Variance	0.98

Mean Centering: $X_{centered} = X - \bar{X}$

Covariance Matrix: $\Sigma = \frac{1}{n-1} (X_{centered})^T X_{centered}$

Eigen Decomposition: $\Sigma v = \lambda v$

Projection: $Z = X_{centered} W$

Symbol	Description
X	Input data matrix
Σ	Covariance matrix
v, λ	Eigenvectors and eigenvalues
W	Projection matrix (top k eigenvectors)
Z	Projected data

Fig 2: PCA Architecture and Hyperparameters

Autoencoders for Feature Reduction

Autoencoder

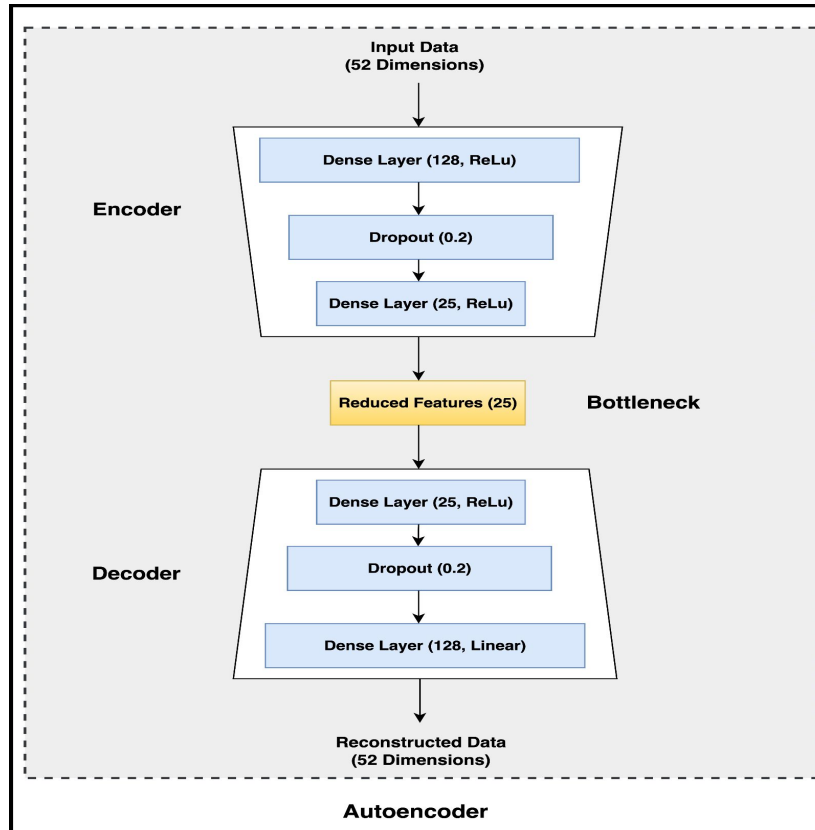
1. Non-Linear
Machine dynamics

2. Task relevant
compression

3. Time awareness

4. Denoising

5. Flexible



Hyperparameters	Value
Scaling	StandardScaler()
Window Size	60
Latent Dim Size	25
Encoder Layers	2 Layers; ReLu Dropout = 0.2
Decoder Layers	2 Layers; ReLu Linear as Output layer Dropout = 0.2
Optimizer	Adam
Learning Rate	0.001
Training Strategy	Early Stopping; Patience = 10
Batch_size	64

Fig 3: Autoencoder Architecture and Hyperparameters ^[2]

LSTM + CNN - Autoencoder for Feature Reduction

LSTM + CNN - Autoencoder

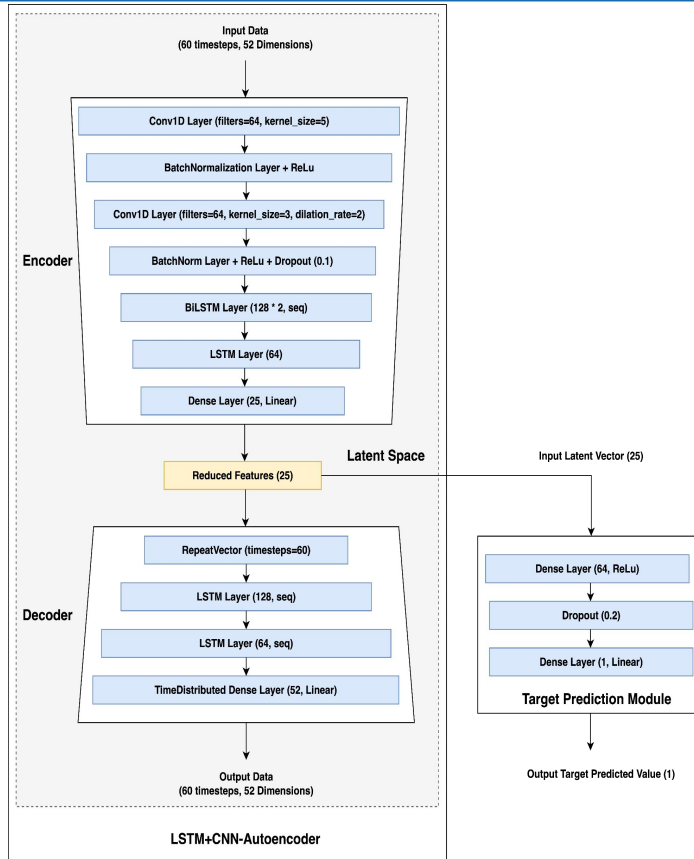
1. Captures Temporal
Dynamics via LSTM

2. Extraction Local
Features through CNN

3. Temporal-Spatial
Fusion with Hybrid
LSTM+CNN

4. Noise Robustness
using CNN filters + LSTM
smoothing

5. Better Generalization
to Complex Machine
States



Hyperparameters	Value
Scaling	StandardScaler()
Window Size	60 (Autoencoder), 10 (Predictor)
Latent Dim Size	25
Encoder Layers	2 Conv1D; ReLu, 2 LSTM (BiLSTM, LSTM), 1 Dense, Dropout(0.1)
Decoder Layers	2 LSTM, 1 RepeatVector (60 timesteps) TimeDistributed Dense; Linear
Regression Layers	2 Dense Layers; ReLu; Linear, Dropout (0.2)
CNN Filters, Kernels, Dilation_rate	Filters = 64, Kernel size 5 and 3, Dilation = 2
Optimizer	Adam
Loss Function	MSE (Reconstruction) + α * MSE (Predictor) $\alpha = 0.5$
Learning Rate	0.001
Training Strategy	Early Stopping; Patience = 10
Batch_size	64

Fig 4: Supervised LSTM+CNN Autoencoder and Hyperparameters ^[1]

Assessing the Quality of Reduced Features

Evaluation

1. Data Input

Reduced Features
Sets

14 (PCA)

25 (Autoencoders)

2. Data Splitting

Training Set = 60%

Validation Set = 20%

Test Set = 20%

3. Data Sequencing

Sliding Windows

Important to learn
Temporal Features

4. LSTM Training

Hyperparameter
Configuration

Monitor Training Loss

EarlyStopping

5. Evaluation Results

Metrics used : MAE,
RMSE, R^2

**Higher R^2 = Reduced
Features **effectively**
capture underlying
data patterns**

Our Results

Evaluation Results

Model Predictions vs Ground Truth

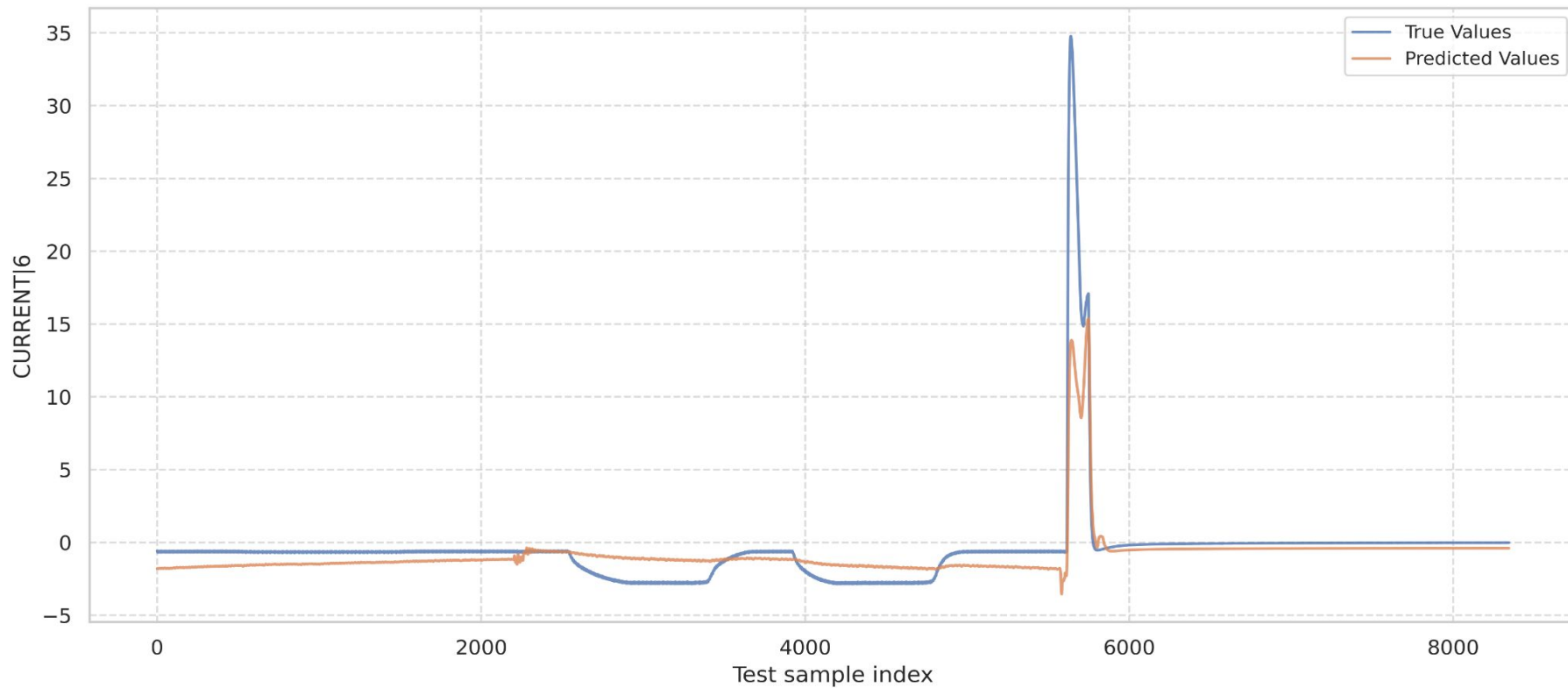


Fig 5: Model Prediction Performance on PCA-Reduced Features

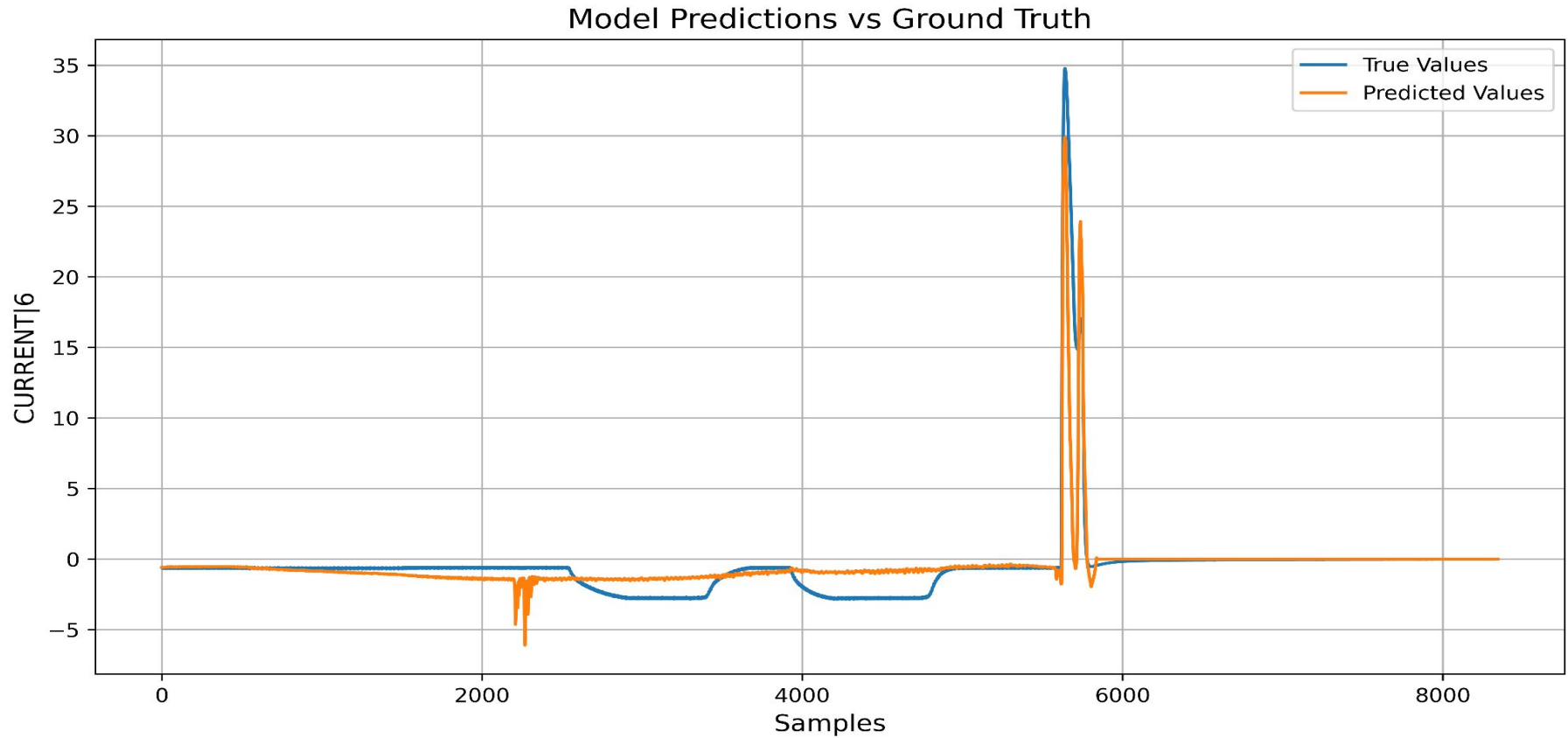


Fig 5: Model Prediction Performance on Autoencoder-Reduced Features

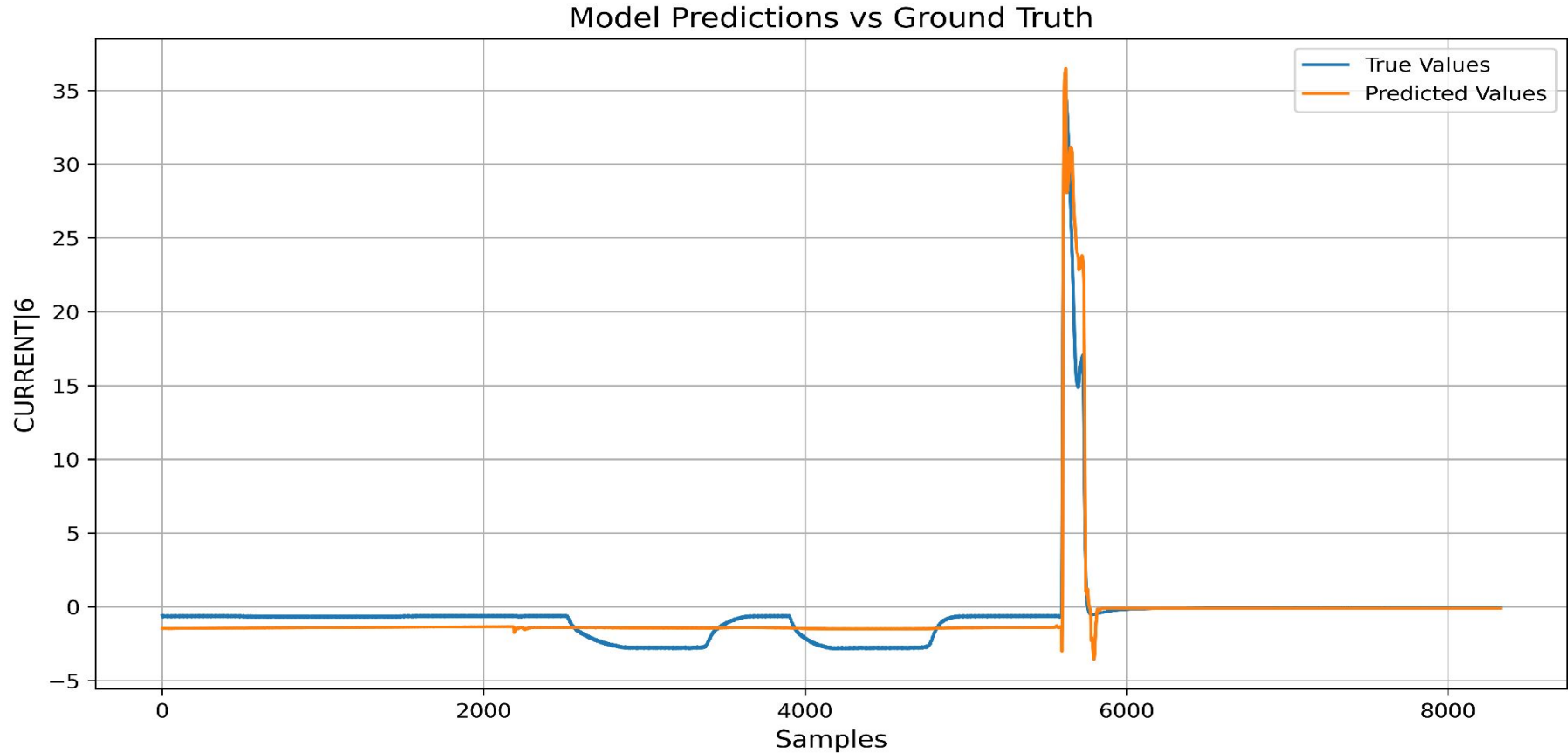


Fig 5: Model Prediction Performance on Autoencoder-Reduced Features

Model	Num. of Features	MAE	RMSE	R ²	Training Time (sec)
Original Dataset	52	0.59	1.79	0.70	200.30
PCA	14	1.15	1.82	0.68	51.41
Autoencoder	25	1.22	2.37	0.70	110.60
LSTM+CNN-Autoencoder	25	0.86	2.38	0.85	121.95

From Black-Box to Transparent Insights

Interpretability

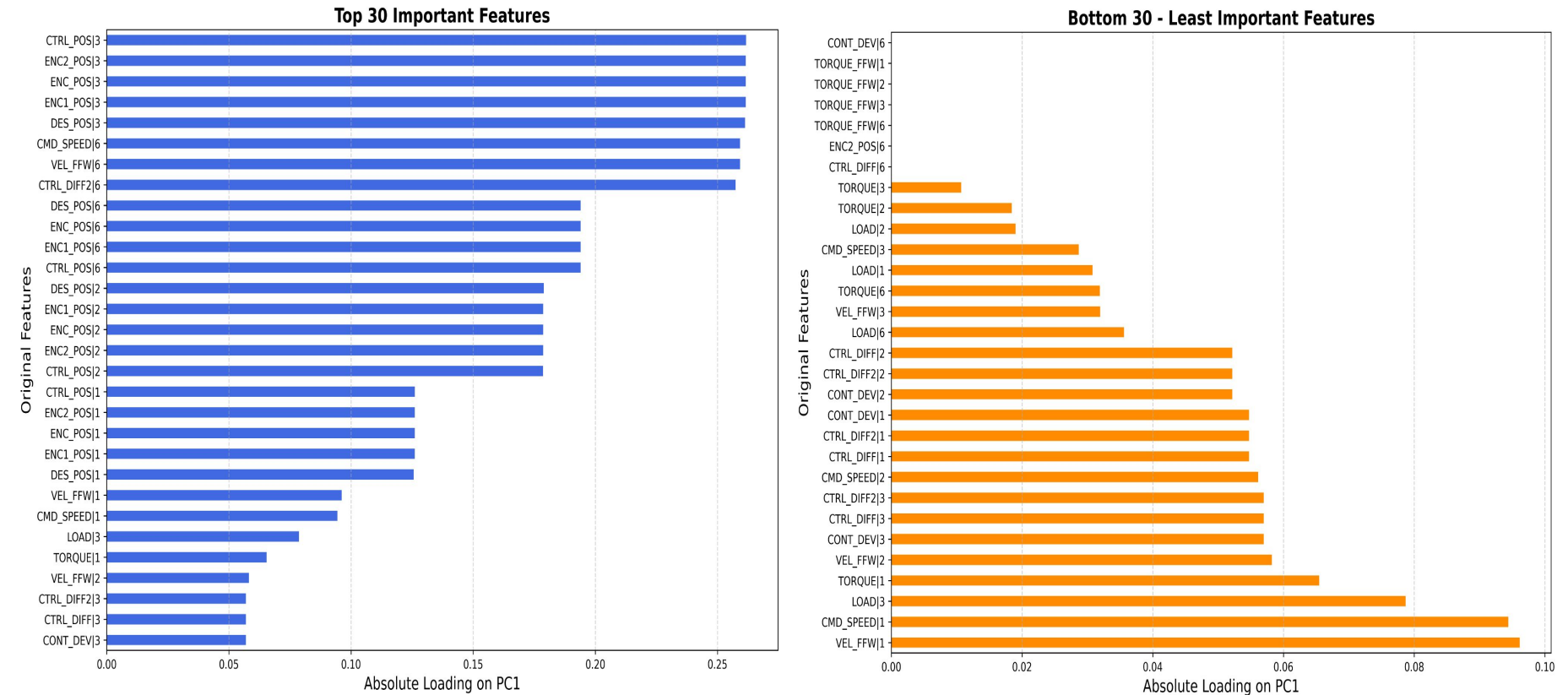


Fig 7: PCA Loadings for Feature Importance

1. Select Explanation Target

Target: Latent Dimension (25)

Explain **which Original Features (52)** influenced most.

2. Prepare Data

Training Subset as Baseline - 300 Windows

Test Windows as explain set. (200 Windows)

Window_Size = 60

3. Initialize Explainer

GradientExplainer

Inputs:
Encoder and Baseline (Training subset)

4. Compute SHAP Values

Run GradientExplainer on Test Subset

Output : SHAP Tensor

Contribution of **Each Feature, at Each Timestep to Each Latent Dim**

Shape:
(Test Windows = 200, Timesteps = 60, Original Features = 52, Latent Dim = 25)

5. Aggregate SHAP Values

Average Absolute SHAP

One Score per Feature (52 total)

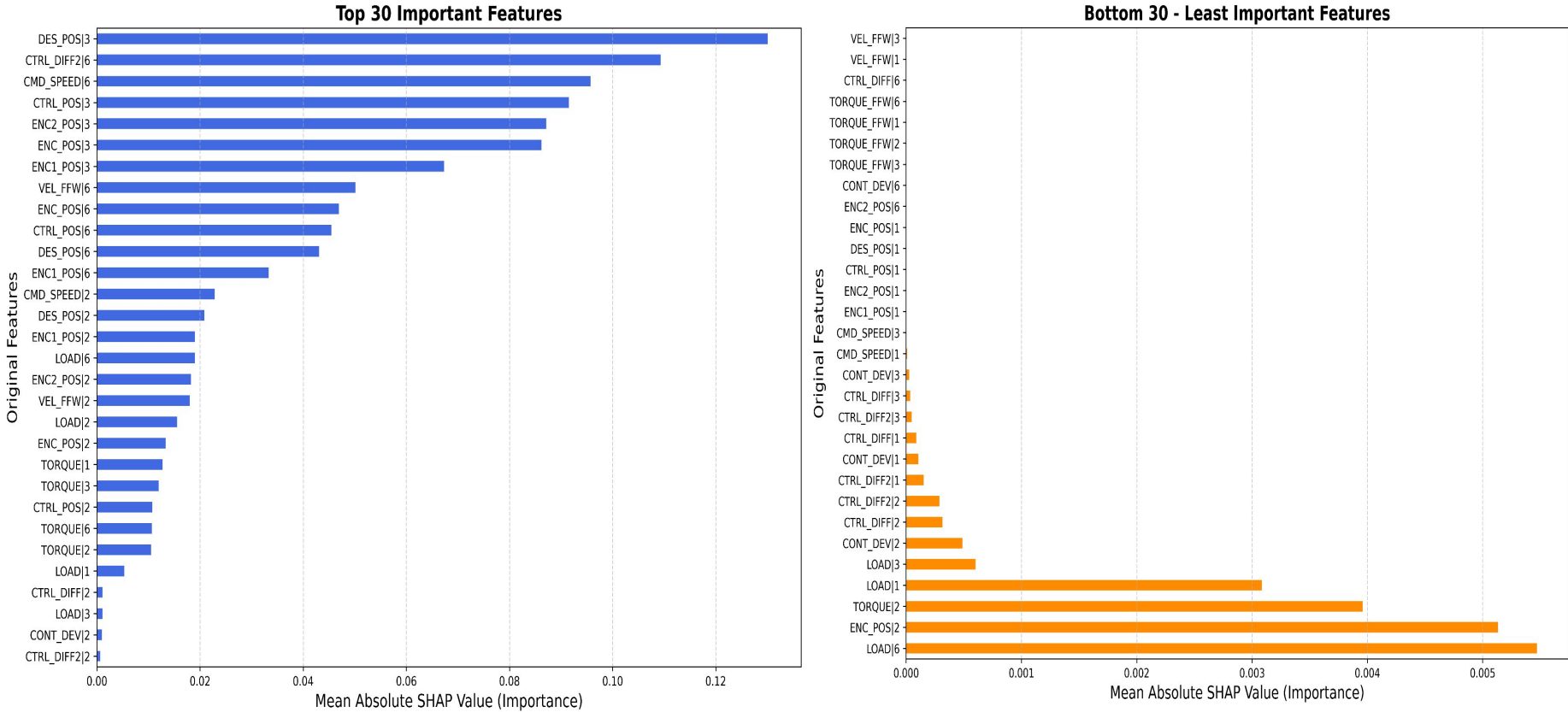
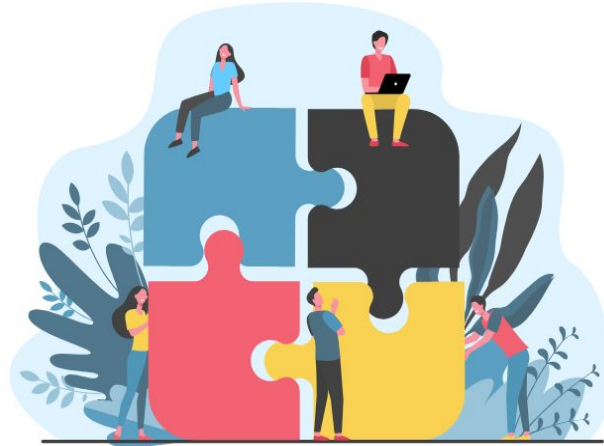


Fig 7: SHAP Scores for Feature Importance

Key Findings and Takeaways

Conclusion



1. **Achieved effective dimensionality reduction while preserving predictive performance** of original features.
2. **Supervised LSTM+CNN Autoencoder** delivered the highest R^2 (~0.80) and **reduced training time by ~40%**.
3. Identified **key contributing** and **redundant** features. enabling potential **cost savings** in sensor deployment and computation.
 - a. Important Features: **CTRL_POS|3, Encoder_POS|3**
 - b. Least Contributing Features: **TORQUE_FFW|1, TORQUE_FFW|2**
4. **Demonstrated that Interpretable Feature Reduction** is possible in Black-Box Autoencoders using Explainable AI.

Opportunities for Further Development and Improvement

Future Work



*TO BE
CONTINUED*

1. To Explore Advanced Dimensionality Reduction Architectures for **smoother Latent Spaces and Better Generalization**. For Ex:
 - a. Variational/Transformer-based Autoencoders
 - b. MultiTask Elastic Net
 - c. Hyperparameter Tuning for better RMSE values.
2. **Enhance Model Interpretability** by validating Feature Importance across multiple XAI techniques
3. Target-level Explainability: Use SHAP on CURRENT|6 Regressor to **Rank Original Features/Sensors driving Energy (Current) usage**

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Thank You For Your Attention!